ChatGPT Upgrade Share

How would you install a package that is not available in the public npm registry?

Fast answer

If a package isn't available in the public npm registry, you have several options depending on where the package is hosted:

1. **Install from a Git repository**

   Bash
   ```bash
   npm install git+https://github.com/username/repo.git
   ```

   Or with SSH:

   Bash
   ```bash
   npm install git+ssh://git@github.com:username/repo.git
   ```

   You can also install a specific branch, tag, or commit:

   Bash
   ```bash
   npm install github:username/repo#main
   npm install github:username/repo#v1.2.0
   npm install github:username/repo#<commit-hash>
   ```

2. **Install from a local folder**

   Bash
   ```bash
   npm install ../my-package
   ```

   Or from an absolute path:

   Bash
   ```bash
   npm install /path/to/my-package
   ```

3. **Install from a local tarball (`.tgz`)**

Bash

```bash
npm install ./my-package-1.0.0.tgz
```

4. **Install from a private npm registry**
   Configure your `.npmrc` file with the registry URL and authentication token:

   INI

   ```ini
   registry=https://your-private-registry.example.com/
   //your-private-registry.example.com/:_authToken=YOUR_TOKEN
   ```

   Then install normally:

   Bash

   ```bash
   npm install package-name
   ```

5. **Install directly from a URL**

   Bash

   ```bash
   npm install https://example.com/packages/my-package-1.0.0.tgz
   ```

### Example (`package.json`)

You can also declare these dependencies directly:

JSON

```json
{
  "dependencies": {
    "my-lib": "github:username/my-lib",
    "local-lib": "file:../local-lib"
  }
}
```

These approaches let you use packages that are hosted privately, stored locally, or maintained in a Git repository instead of the public npm registry.

Today 8:39 AM

How many ways you can hosted privately a library or stored locally or maintained in a Git repository instead of the public npm registry?

In enterprise environments, there are **7 common ways** to host or consume a package without publishing it to the public npm registry.

| Method | Best For | Used in Enterprise? | Example |
|---|---|---|---|
| **1. Local Folder** | Development/testing | ✅ Yes | `npm install ../shared-lib` |
| **2. Local Tarball (.tgz)** | Offline installation, sharing builds | ✅ Yes | `npm install ./my-lib-1.0.0.tgz` |
| **3. Git Repository (GitHub/GitLab/Bitbucket/Azure Repos)** | Small internal libraries | ✅ Yes | `npm install github:user/lib` |
| **4. Private npm Registry** | Enterprise package management | ⭐⭐⭐ Most Common | Nexus, Verdaccio, Artifactory |
| **5. Cloud Package Registry** | Cloud-native organizations | ⭐⭐⭐ Very Common | GitHub Packages, AWS CodeArtifact, Azure Artifacts, Google Artifact Registry |
| **6. Direct URL (HTTP/HTTPS)** | Temporary distribution | Rare | `npm install https://company.com/lib.tgz` |
| **7. npm Link / Workspace / Monorepo** | Local development across projects | ⭐⭐⭐ Very Common | `npm link`, npm workspaces, Nx, Turborepo |

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## 1. Local Folder

```
ProjectA
ProjectB
```

Bash

```bash
npm install ../ProjectB
```

or

```json
{
  "dependencies": {
    "shared-lib": "file:../shared-lib"
  }
}
```

**Use case**

- Developing two projects simultaneously
- Local testing

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## 2. Tarball (.tgz)

Create package

```bash
npm pack
```

Produces

```
my-lib-1.0.0.tgz
```

Install

```bash
npm install ./my-lib-1.0.0.tgz
```

Used when

- No internet
- Air-gapped environments
- Sharing internal builds

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## 3. Git Repository

Can be hosted in

- GitHub

- GitLab
- Bitbucket
- Azure DevOps
- Self-hosted GitLab

Examples

Bash

```bash
npm install git+https://github.com/company/shared-lib.git
```

or

Bash

```bash
npm install github:company/shared-lib
```

Specific branch

Bash

```bash
npm install github:company/shared-lib#develop
```

Specific tag

Bash

```bash
npm install github:company/shared-lib#v2.0.1
```

Specific commit

Bash

```bash
npm install github:company/shared-lib#4ef23ab
```

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## 4. Private npm Registry (Most Common)

Companies usually host internal registries such as:

- Sonatype Nexus Repository
- JFrog Artifactory
- Verdaccio
- npm Enterprise (legacy)

Example

```
Developer
     |
     ▼
npm install company-ui
     |
     ▼
Private Registry
     |
     ▼
Company Package
```

`.npmrc`

INI

```ini
registry=https://npm.company.com/
//npm.company.com/:_authToken=xxxxx
```

Install

Bash

```bash
npm install @company/ui
```

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## 5. Cloud Package Registry

Examples

- GitHub Packages
- AWS CodeArtifact
- Azure Artifacts
- Google Artifact Registry

Example

INI

```ini
@company:registry=https://npm.pkg.github.com
```

Install

Bash

```bash
npm install @company/ui
```

Common in cloud-based organizations.

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## 6. Direct URL

Package hosted on a server

```bash
npm install https://company.com/packages/ui-2.0.0.tgz
```

Useful for temporary distributions or CI/CD artifacts.

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## 7. npm Link / Workspaces / Monorepo

### npm link

```bash
cd shared-lib
npm link

cd app
npm link shared-lib
```

Changes are reflected immediately.

### npm Workspaces

```
root/
   package.json

packages/
   ui/
   api/
   auth/
```

```bash
npm install
```

npm links all workspace packages automatically.

### Nx / Turborepo

Large organizations commonly use monorepo tools to manage multiple applications and shared libraries in a single repository with consistent dependency management.

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## Which methods do enterprises actually use?

| Method | Popularity |
|---|---|
| Local Folder | ⭐⭐ |
| Tarball | ⭐ |
| Git Repository | ⭐⭐⭐ |
| Private Registry | ⭐⭐⭐⭐⭐ |
| Cloud Package Registry | ⭐⭐⭐⭐⭐ |
| Direct URL | ⭐ |
| npm Workspaces / Monorepo | ⭐⭐⭐⭐⭐ |

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### Real-world examples

- **Google**: Monorepo with internal build/package infrastructure (not public npm).
- **Microsoft**: Azure Artifacts and internal package feeds.
- **Amazon (AWS)**: AWS CodeArtifact and internal repositories.
- **Netflix**: Internal registries and monorepo tooling.
- **Banking/Financial institutions**: Commonly use **JFrog Artifactory**, **Sonatype Nexus**, or **Verdaccio** hosted within their private networks for security and governance.

For interview purposes, it's worth emphasizing that **private registries (Artifactory, Nexus, Verdaccio, Azure Artifacts, AWS CodeArtifact)** and **Git-based dependencies** are the most common alternatives to the public npm registry in enterprise environments.

create an image